

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** MENA Multi-Country Arabic Tutoring System — Eight-Country Curriculum Deployment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content (label correctness for the deployment context) is the dimension most exposed to validity violations. The 96% spot-check agreement **[Q43, Q44]** was conducted on 100 randomly sampled questions **[Q42]** without country stratification, so it cannot speak to per-country annotation quality. Annotators are exclusively from Jordan, Egypt, Lebanon, UAE, and KSA **[Q31]** — Morocco, Kuwait, and Palestine have no documented annotators **[Q96]**, yet Palestine is a top-three question contributor **[WEB-1]** and Moroccan law content is present **[DATASET-D34]**. This means Palestinian, Moroccan, and Kuwaiti questions (where present) had ground-truth labels verified by workers from other national educational contexts. Empirically, the dataset analysis shows: (a) Jordanian-curriculum facts (King Talal's reign, 1952 constitution, Royal Medical Services, CBJ URL) are encoded as universally correct answers **[DATASET-D3, D4, D7, D20, D21]**; (b) Egypt-specific currency and institutions are encoded as universal in university economics **[DATASET-D28, D39, D40]**; (c) Moroccan procedural law is encoded as 'the' law answer without jurisdiction tag **[DATASET-D8, D9, D10, D34]**; (d) at least one likely-incorrect ground-truth label is observed **[DATASET-D32]**; (e) OCR-corrupted questions **[DATASET-D14, D15]** cannot have reliably verifiable answer keys. Pretraining contamination is also acknowledged as not ruled out **[Q94, Q95]**. For a deployment that requires curriculum-aligned answers per country [elicitation A1] and country-specific legal correctness [elicitation A2], the ground-truth labels do not reliably reflect regional stakeholder perspectives for Moroccan, Emirati, Kuwaiti, Saudi, Lebanese, or Palestinian cohorts.

ID	Evidence
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
Q94	"It is important to emphasize that our experimental results do not provide conclusive answers regarding the performance of LLMs in Arabic." (p.9)
Q95	"to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)
WEB-1	Models perform notably poorly on Morocco- and UAE-sourced questions, consistent with thin annotation coverage (https://arxiv.org/html/2402.12840v1)
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D3	"بوهة فاك ةلودل نوؤش لىل ةماعل ةيالول بحاص سلجمل" with answer "ارازول سلجمل" — "The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure
D28	١٠٠٠٠٠ بولطم الامثتسالا جج ناك اذا — ١٥ % بوهة رمثتسم الامل لاولال لىل ةماعل لدعم...هينج — Egyptian currency (جني = Egyptian pound) in university economics — Egypt-specific monetary context

D8	— يوعدلا رهوج يف مكحل رودص دعب الا عوفد وأ ضراع عازن يف ةرداصل وأ ةيدي مهمتلا ماكحلأ فانئتسا لبقي ال Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D32	تايمكب هرفاوت: لىل لك ذى ببسلا عجرى...يغبطل زاغلاو طفنلا نم ةربكب تاورثب ةيربعل اندالبل لىل علل ملل معنأ "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science
D14	Severely — لىل ريذت...ةتفلتخسل ةتيرادلإا تايتدتسل او صاختلاأل نيبت ةطلتدل ةتيتيت ةتجرد OCR-distorted text about authority distribution in organizations, from Egyptian source
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom — ةكلمملا يف ةيرملا تاداعلل اقبط نوكت نا اهتسرامم روتسدل طرتشي يتلا تايرحلل وقوقحل نم
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact — ةيمسرلا ةيلااملا تامدخل لىل لوصولو نوعيطتسي ال نذل نيي ندرأل ةبسن
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services — ةيبطل ةني دمل ايفشتسم: اهنم...تاسسؤملا نم ديدعل ةيموكحل ةيكلملا ةيبطل تامدخل لمشت
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school — ةنس همكح دموم، 1952: يف كلذو ةيحص بابسلأل هتياوالت تهنتا ني سحل نب لال طلكملا
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
D40	Source: http://www.aun.edu.eg — all 5 from same Assiut University Egypt URL — University economics exclusively from Egyptian university source
D9	عافدل اب ةفل كملا ةيموكحل اطلسل او لدعل ربزو نم راربب...نيي كيردلل ةيئاضقلل ططرشلل طباض ةفص لي وخت نكمي Moroccan judicial police powers — references "يكلمل كرل" (Royal Gendarmerie of Morocco)
D10	Moroccan civil party withdrawal in criminal proceedings — ةيحص ةبوجلأا عيجم: مكحل رودص لبقي ندمل فرطل لزانن اذإ
D15	Severely OCR-corrupted management question with unreadable answer options — ت. ح... مح رل...مادختسلإا ةسئاد طخل لىل علل اثم دعي ل

Strengths

- Annotator credentials are documented: Master's students/Research Assistants and Bachelor's-degree-holders, all native MSA speakers [Q35, Q39]
- 96% inter-annotator agreement on 100-question spot check provides a baseline quality signal [Q43]
- Compensation exceeded monthly average wages in respective countries [Q36]
- One-hour orientation workshop conducted before data collection [Q40]
- STEM ground truth (math, biology, physics, factual CS) is curriculum-neutral and reliable across all eight countries [DATASET-D24, strength 5]

ID	Evidence
Q35	"Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)
Q39	"While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q36	"We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)
Q40	"Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)
D24	إذإ ؟ اهتيتبثت مت يتلا (CO2) تائي زج ددع مكف نفا لك ةقلح نع زوكولغ تائي زج (3) جتن اذإ Palestine-sourced biology; universal factual content

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?

OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect regional stakeholder perspectives unevenly. For STEM and curriculum-neutral content, labels are reliable across all eight countries. For civics, history, social science, economics, and law, labels reflect the originating exam's jurisdiction without country-stratified validation [DATASET-D1, D7, D34].

OC-2: Disagreement is highly likely between original annotators and regional populations for: (a) Moroccan students reviewing Jordanian civics [DATASET-D1, D2]; (b) UAE/Kuwait students reviewing Egyptian university economics [DATASET-D28]; (c) Egyptian/Saudi/Jordanian/UAE/Kuwaiti/Lebanese/Palestinian law students reviewing Moroccan procedural law [DATASET-D8, D9, D10]; (d) Palestinian students reviewing Jordanian-narrative King Abdullah I content [DATASET-D31].

OC-3: Annotator demographics documented [Q31] but limited: 10 annotators total, all from Jordan/Egypt/Lebanon/UAE/KSA. No Morocco/Kuwait/Palestine representation [Q96]. No formal datasheet or data statement beyond paper text.

OC-4: Re-annotation by representative annotators from Morocco, Kuwait, and Palestine is essential before deployment use, especially for civics, history, law, social science, and economics subsets.

OC-5: Aggregation: 96% agreement reported as a single global figure [Q43] without country stratification — minority-country perspectives are erased in the aggregate. No formal IAA statistic beyond this [Q43].

OC-6: Documented label issues: (a) Jordanian-curriculum facts as universal ground truth [DATASET-D1, D3, D4, D7, D20, D21]; (b) Egyptian university content as universal [DATASET-D28, D39, D40]; (c) Moroccan-only law ground truth [DATASET-D34]; (d) no Palestinian annotator review of Palestinian content [Q96, WEB-1]; (e) at least one likely-incorrect answer key observed [DATASET-D32]; (f) OCR-corrupted questions where verifiability is compromised [DATASET-D14, D15]; (g) potential pretraining contamination not ruled out [Q94, Q95].

ID	Evidence
D1	"What are the manifestations of change in modern Jordanian society?" — country-specific Jordanian civics
D7	0.67 — "67% of Jordanians cannot access formal financial services" — Jordan-specific statistic as universal economic fact
D34	All 5 Law (Professional) examples: Country = Morocco, from single Google Drive PDF — All sampled law questions are exclusively from Morocco — single-source PDF
D28	10% — Egyptian currency (جنيح = Egyptian pound) in university economics — Egypt-specific monetary context
D8	— Moroccan procedural law — appeals rules under Moroccan Code of Civil Procedure
D31	King Abdullah I's positions on Zionism — highly sensitive political content, Jordanian framing of Palestinian history
D2	"The first coalition government in Jordan in 1952 was known as the ___ government" — Jordanian political history
D9	— Moroccan judicial police powers — references "الملك لكرجل" (Royal Gendarmerie of Morocco)
D10	— Moroccan civil party withdrawal in criminal proceedings
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
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D32	تايكمكب هرفاوت: لىل ك ل ذ ي ف ب ب س ل ع ج ر ي... ي ع ي ب ب ط ل ز ا غ ل و ط ف ن ل ن م ق ر ي ب ك ت ا و ر ث ب ق ي ب ر ع ل ا ن د ا ل ب ل ع ل ل ل م ع ن ا "Gas is available in limited quantities" — possibly incorrect answer key for general Arab world social science
D14	Severely OCR-distorted text about authority distribution in organizations, from Egyptian source ...ت ي ر ا د ل ا ت ي ر ا د ل ا ت ي ه ت د س ل و ص ا خ ت ل ا ل ن ي ت ب ط ل ت د ل ع ت ي ي ه ت ع ت ج ر د
D3	"The council with general sovereignty over all state affairs is the Council of Ministers" — specifically Jordanian constitutional structure ب و ه ة ف ا ك ة ل و د ل ن و و ش ل ع ل ع ا م ا ل ع ي ا ل و ل ب ح ا ص س ل ج م ل
D4	"Rights under the [Jordanian] Kingdom's constitution" — explicitly references the Jordanian Hashemite Kingdom م ا ه ت س ر ا م م ر و ت س د ل ط ر ت ش ي ي ت ل ت ا ي ر ح ل و ا ل ق و ق ح ل ن م
D20	"Royal Medical Services include: Medical City Hospital" — specifically Jordanian Royal Medical Services م ا ه ن م... ت ا س س و م ل ن م د ي د ع ل ا ي م و ك ح ل ا ي م ل م ل ا ي ب ط ل ا م د خ ل ل م ش ت
D21	"King Talal bin Hussein's reign ended for health reasons in 1952 — one year" — Jordanian royal history in primary school ي ف ك ل ذ و ق ي ح ص ب ا ب س ا ل ه ت ي ا ل و ت ه ت ن ن ي س ح ل ن ب ل ا ل ط ل م ل
D39	Source: http://www.aun.edu.eg/commerce/... — All 5 accounting examples from Assiut University (Egypt) — single Egyptian university source
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D15	Severely OCR-corrupted management question with unreadable answer options ت ح ل ... ت م ح ر خ ر ل ... م ا د خ ت س ل ا ة س ن ا د ط ط خ ل ل ع ل ا ل ا ث م د ع ي ل ا

Information Gaps

- Country-stratified inter-annotator agreement is not reported [Q43]
- Pretraining contamination extent is not measured [Q95]
- Total proportion of incorrect answer keys beyond the 4% noise floor implied by 96% agreement is unknown

Requires Expert Verification

- Per-country curriculum experts (especially Moroccan Maliki Islamic studies, Kuwaiti civics, Palestinian national history, UAE 4-4-4 curriculum) to re-validate ground-truth labels
- Legal experts from Egypt, UAE, KSA, Jordan, Lebanon, Kuwait, Palestine to assess law-subset jurisdictional correctness

Remediation

Gap 1: Ground-truth labels for Palestinian, Moroccan, and Kuwaiti questions were verified by workers without those national curriculum backgrounds [Q31, Q96]; 96% agreement is a global figure not stratified by country [Q43].

Recommendation: Recruit at least 2–3 curriculum experts each from Morocco, Kuwait, and Palestine to re-verify ground-truth labels for at least the civics, history, social science, and law subsets relevant to those countries. Report country-stratified IAA before deployment use.

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